

McCulloch and Till's combination of skills, in cellular biology and biophysics, proved powerful. In 1961, while carrying out research into bone marrow cells, the two discovered the first clue to the existence of a special type of cell that became known as the stem cell. These cells have a unique potential to become other, specialized types of cells and can also self-renew and multiply themselves.

Unexpected developments

The game-changing discovery came about quite by chance; McCulloch described it as a case study in "the importance of serendipity in scientific research." McCulloch was interested in the medical potential of nuclear radiation and wanted to explore its effects on cancer cells. His project involved irradiating mice, and McCulloch, a medical doctor, needed a physicist with the necessary expertise in radiation—Till.

The two began measuring the radiation sensitivity of the mice's bone marrow cells by exposing the mice to radiation that would be lethal if they did not receive a transplant, then replacing their bone marrow cells with donor cells. At the time, bone marrow transplantation was a new treatment known to replenish crucial blood cells. However, Till and McCulloch had also exposed the donor marrow in their transplants to varying doses of radiation. By monitoring the mice's response to the transplants, the scientists could measure how many of the donor cells had survived the differing radiation doses, and therefore the number of live marrow cells the mice had received.

The team repeated the experiment and this time carried out autopsies on the mice 10 days after the transplants.

“Major breakthroughs in science involve unexpected findings. You can’t expect the unexpected.”

James Till, 2018

One Sunday morning, McCulloch went into the laboratory to assess the tissue samples from the autopsies and was surprised to see a number of small bumps on the dead mice's spleens. The number of bumps, or nodules, was in exact proportion to the number of live marrow cells the mice had received.

Colonies of clones

Till and McCulloch needed to find out whether each of these nodules had grown from a single cell or from many cells. So, with Till's PhD student Andrew Becker, they again irradiated the donor cells with the aim of creating a marker, or trace. In a few of the irradiated donor cells, the radiation caused small changes to the chromosomes. These altered chromosomes acted as a marker, in that they would be present in any new cells that had grown from them. After isolating and examining hundreds of cells from a new set of spleen nodules, the team found that each nodule had grown from a single cell. In other words, they were clones. Till and McCulloch referred to the original cells as "colony-forming units."

More remarkably, the colonies of cells in each nodule were a mixture of those that develop into the three types of blood cells: red cells, white cells, and platelets. Till and McCulloch knew this was exciting, but they were cautious. In 1961, they published their findings in a plainly titled